

Part A: Numerical Expressions and Order of Operations

Evaluate each expression completely. Show each major order-of-operations step.

1. $8 + 8 \cdot 3$

6. $(-6 + (-17))^2 \div (-1)$

2. $5 + 8 \cdot (-8)$

7. $-16 - 3 \cdot ((-9) + (-16))$

3. $-18 \div (-3) - (-7)$

8. $2 - 9 \cdot ((-6) + 2)$

4. $5 + (-5) \cdot (-8)$

9. $6^2 - 7 \cdot 7$

5. $(-4 + 16)^2 \div 8$

10. $(-3)^3 - 17 \cdot 9$

Part B: Properties of Addition and Multiplication

Select the property that is displayed in the given equation.

1. $b \cdot 5 = 5 \cdot b$

- A) Identity Property of Multiplication
- B) Inverse Property of Multiplication
- C) Commutative Property of Multiplication
- D) Associative Property of Multiplication

2. $0 + 5 = 5 + 0$

- A) Commutative Property of Addition
- B) Associative Property of Addition
- C) Inverse Property of Addition
- D) Identity Property of Addition

3. $(3 + 5) + 4 = 3 + (5 + 4)$

- A) Identity Property of Addition
- B) Associative Property of Addition
- C) Commutative Property of Addition
- D) Inverse Property of Addition

4. $(2 + 7) + 1 = 2 + (7 + 1)$

- A) Associative Property of Addition
- B) Inverse Property of Addition
- C) Identity Property of Addition
- D) Commutative Property of Addition

5. $10 \cdot 1 = 10$

- A) Associative Property of Multiplication
B) Inverse Property of Multiplication
C) Identity Property of Multiplication
D) Commutative Property of Multiplication
6. $-10 \cdot 1 = -10$
A) Associative Property of Multiplication
B) Inverse Property of Multiplication
C) Commutative Property of Multiplication
D) Identity Property of Multiplication
7. $-13 \cdot \frac{1}{-13} = 1$
A) Commutative Property of Multiplication
B) Identity Property of Multiplication
C) Inverse Property of Multiplication
D) Associative Property of Multiplication
8. $14 + (-14) = 0$
A) Identity Property of Addition
B) Inverse Property of Addition
C) Associative Property of Addition
D) Commutative Property of Addition
9. $1 \cdot y = y$
A) Commutative Property of Multiplication
B) Associative Property of Multiplication
C) Identity Property of Multiplication
D) Inverse Property of Multiplication
10. $x + b = b + x$
A) Identity Property of Addition
B) Inverse Property of Addition
C) Commutative Property of Addition
D) Associative Property of Addition

Part C: Evaluating Expressions by Substitution

Substitute the given value or values and evaluate completely.

1. $b + 2b + 3b - 3q^2 + 5$, $b = -3$, $q = -6$ 4. $-3b - 3b - 2b - 3q^2$, $b = 3$, $q = -1$
2. $-2p - p + 2q^2 + 9$, $p = -8$, $q = 8$ 5. $-4c^2 + 2c + 3p - 4$, $c = 2$, $p = -6$
3. $4h^2 - 3h + 4h - q + 1$, $h = -6$, $q = -5$

6. $f(x) = 2x^2 + 5x + 3$, $f(3)$

9. $f(x) = x^2 - 6x - 4$, $f(7)$

7. $f(x) = x^2 + x - 1$, $f(2)$

10. $f(x) = 2x^2 - 9x - 9$, $f(4)$

8. $f(x) = 4x^2 - 3x - 3$, $f(-6)$

Part D: Distributing and Combining Like Terms

Rewrite each expression in simplest terms. Show the distribution step.

1. $9x - 1 + 4x + 2$

8. $4(3x - 11) - 6x + 6$

2. $x + 1 - 6x + 7$

9. $4(5x + 8) + 7x - 8$

3. $x + 2 - 8x + 6$

10. $-3(2x - 1) - 9x + 12$

4. $8x + 5 + 7x + 5$

11. $(5x + 8) - 2x + 1$

5. $4x + 10x^2 + 5x^2 + 5x$

12. $4(9x + 2) + 10x - 7$

6. $4x^2 + 7x + 8x + 12x^2$

13. $9(-5x + 2) - 8x + 11$

7. $12x - 8x^2 + 12x^2 + 2x$

14. $-12(11x + 9) + 4x - 1$

Part E: Absolute Value

Evaluate each expression. Work inside the absolute value bars first.

1. $|12 - 14| - 3| - 8 - 8|$

3. $|7 - 10| - |13 - 10|$

2. $|-4 - 3| - |-2 + 1|$

4. $|2 - 13| - |13 + 13|$

5. $|-3 + 5| - 3| - 6 - 7|$

8. $|2 + 8| + 3|1 + 6|$

6. $|-3 + 13| + 2| - 1 - 8|$

9. $|-7 + 13| + 3|4 + 6|$

7. $|-7 - 14| + 2| - 10 + 6|$

10. $|2 + 9| + |2 - 14|$

Part F: Fraction Arithmetic

Evaluate. Write every answer in lowest terms.

1. $1 + \frac{5}{9}$

5. $\frac{1}{6} \cdot \frac{2}{5}$

2. $\frac{1}{2} + \frac{5}{8}$

6. $\frac{1}{6} \div 3\frac{2}{3}$

3. $\frac{2}{5} - 2$

7. $\frac{10}{6}$

4. $1\frac{1}{3} \cdot \frac{1}{6}$

8. $\frac{3}{6}$

Part G: One- and Two-Step Equations

Solve for the variable and simplify your answer.

1. $x + 9 = 4$

5. $x - 4 = 4$

2. $4x = 36$

6. $7x + 9 = \frac{1}{4}$

3. $4x = 28$

7. $3x + 8 = \frac{5}{4}$

4. $x + 4 = 2$

8. $8x + 6 = -90$

11. $9x - 10 = 14$

9. $9x + 2 = -61$

12. $8x - 1 = 39$

10. $11x - 3 = \frac{68}{3}$

Part H: Multi-Step Equations with Distribution*Distribute first, then solve for the variable.*

1. $4(6x + 12) - 5 = 307$

6. $7(7x - 11) - 2 = -520$

2. $5(7x + 10) - 3 = 82$

7. $6(2x - 12) + 6 = 54$

3. $5(2x - 8) + 6 = -74$

8. $6(x - 5) + 6 = -66$

4. $4(9x - 3) + 6 = 210$

9. $7(5x + 4) - 2 = 236$

5. $3(8x - 3) - 12 = 99$

10. $6(7x - 4) - 11 = 259$

Part I: Variables on Both Sides*Solve for the variable. Clear fractions first where it helps.*

1. $-2x + 8 = -4x + 5$

3. $6x + 5 = x + \frac{35}{2}$

2. $4x + 9 = 12x - 87$

4. $-12x + 8 = -5x + 92$

5. $-10x - 6 = 10x - 166$

11. $\frac{x}{4} + 18 = \frac{67}{4}$

6. $8x - 6 = -6x - 13$

12. $\frac{2}{3}x - 19 = -\frac{56}{3}$

7. $6x + 9 = -x - 33$

13. $\frac{x + 20}{2} + 12 = \frac{62}{3}$

8. $9x - 3 = 2x - 10$

14. $\frac{x}{4} + 4 = \frac{33}{4}$

9. $-7x - 5 = -8x - \frac{25}{4}$

15. $\frac{x}{4} - 6 = -\frac{33}{8}$

10. $x - 1 = -11x + 107$

16. $\frac{1}{6}x - 16 = -\frac{143}{9}$

Answer Key

Part A: Numerical Expressions and Order of Operations

- | | | |
|--------|---------|----------|
| 1. 32 | 5. 18 | 9. -13 |
| 2. -59 | 6. -529 | 10. -180 |
| 3. 13 | 7. 59 | |
| 4. 45 | 8. 38 | |

Part B: Properties of Addition and Multiplication

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|--|---|---|
| 1. C) Commutative Property of Multiplication | 5. C) Identity Property of Multiplication | 8. B) Inverse Property of Addition |
| 2. A) Commutative Property of Addition | 6. D) Identity Property of Multiplication | 9. C) Identity Property of Multiplication |
| 3. B) Associative Property of Addition | 7. C) Inverse Property of Multiplication | 10. C) Commutative Property of Addition |
| 4. A) Associative Property of Multiplication | | |

Part C: Evaluating Expressions by Substitution

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|---------|--------|---------|
| 1. -121 | 5. -34 | 9. 3 |
| 2. 161 | 6. 36 | 10. -13 |
| 3. 144 | 7. 5 | |
| 4. -27 | 8. 159 | |

Part D: Distributing and Combining Like Terms

- | | | |
|-----------------|------------------|-------------------|
| 1. $13x + 1$ | 6. $16x^2 + 15x$ | 11. $3x + 9$ |
| 2. $8 - 5x$ | 7. $4x^2 + 14x$ | 12. $46x + 1$ |
| 3. $8 - 7x$ | 8. $6x - 38$ | 13. $29 - 53x$ |
| 4. $15x + 10$ | 9. $27x + 24$ | 14. $-128x - 109$ |
| 5. $15x^2 + 9x$ | 10. $15 - 15x$ | |

Part E: Absolute Value

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|--------|--------|--------|
| 1. -46 | 5. -37 | 9. 36 |
| 2. 6 | 6. 28 | 10. 23 |
| 3. 0 | 7. 29 | |
| 4. -15 | 8. 31 | |

Part F: Fraction Arithmetic

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|--------------------|-------------------|-------------------|
| 1. $1\frac{5}{9}$ | 4. $\frac{2}{9}$ | 7. $1\frac{2}{3}$ |
| 2. $1\frac{1}{8}$ | 5. $\frac{1}{15}$ | 8. $\frac{1}{2}$ |
| 3. $-1\frac{3}{5}$ | 6. $\frac{1}{22}$ | |

Part G: One- and Two-Step Equations

1. $x = -5$

2. $x = 9$

3. $x = 7$

4. $x = -2$

5. $x = 8$

6. $x = -\frac{5}{4}$

7. $x = -\frac{9}{4}$

8. $x = -12$

9. $x = -7$

10. $x = \frac{7}{3}$

11. $x = \frac{8}{3}$

12. $x = 5$

Part H: Multi-Step Equations with Distribution

1. $x = 11$

2. $x = 1$

3. $x = -4$

4. $x = 6$

5. $x = 5$

6. $x = -9$

7. $x = 10$

8. $x = -7$

9. $x = 6$

10. $x = 7$

Part I: Variables on Both Sides

1. $x = -\frac{3}{2}$

2. $x = 12$

3. $x = \frac{5}{2}$

4. $x = -12$

5. $x = 8$

6. $x = -\frac{1}{2}$

7. $x = -6$

8. $x = -1$

9. $x = -\frac{5}{4}$

10. $x = 9$

11. $x = -5$

12. $x = \frac{1}{2}$

13. $x = -\frac{8}{3}$

14. $x = 17$

15. $x = \frac{15}{2}$

16. $x = \frac{2}{3}$